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Microcontrollers Reflection - Sensing The World

During this module, I learned about microcontrollers and how they can be used to control electronic devices. My group and I worked with an Arduino Nano ESP32 and learned how to write code using MicroPython. We built simple circuits on a breadboard using components such as resistors and LEDs. To complete the activities, we followed circuit diagrams, connected the components on the breadboard, and uploaded code to the microcontroller to control the behavior of the LEDs. We also used AI tools to help generate and troubleshoot code for our DIY circuits. Working in a group allowed us to divide tasks, compare ideas, and solve problems together when our circuits or code did not work as expected. Additionally, I learned how to use GitHub, which allows me to store all of my codes and keep them organized.

Through this experience, I gained a better understanding of how hardware and software work together in electronic systems. I also learned the importance of resistors in protecting LEDs and controlling the flow of electricity within a circuit. One of the most interesting parts of the module was using AI to help write code because it showed me how technology can be used as a tool to support learning and problem-solving. At times, troubleshooting the circuits was challenging because a small wiring mistake could prevent the entire project from working. However, those challenges helped me develop patience and debugging skills. Overall, this module was a valuable experience that improved my understanding of electronics, programming, and teamwork while giving me hands-on experience building and testing real circuits. Throughout this week, I also worked on my GitHub portfolio. I added links to my LinkedIn, resume, and social media, so that future employers/peers can view my work.